

1. $A = 3x^2 + 4x + 3$, $B = 2x^2 - 4x - 4$, $C = 2x^2 - 2x - 5$ であるとき，次の計算をせよ。

(1) $B + C$

答.

(2) $A - 3C$

答.

(3) $A - B - C$

答.

(4) $-A + B - C$

答.

(5) $-A - 2B + 3C$

答.

(6) $-3A - 2B + C$

答.

(7) $A + 2B - 3C$

答.

(8) $2A - 3B + 3C$

答.

2. 次の 1 次方程式を解け。

(1) $5x - 5 = 4x - 5$

答.

(2) $5x - 4 = -2x + 3$

答.

(3) $-3x + 3(6x - 7) = 5$

答.

(4) $5x + 8 - (4x + 4) = 0$

答.

(5) $2x - (5x + 6) = 4$

答.

(6) $-\frac{9}{4}x - 1 = 3x + 1$

答.

(7) $\frac{9}{2}x - 9 = -6x + \frac{7}{4}$

答.

(8) $-5x - \frac{1}{4} = 2x + \frac{7}{4}$

答.

1. $A = 3x^2 + 4x + 3$, $B = 2x^2 - 4x - 4$, $C = 2x^2 - 2x - 5$ であるとき, 次の計算をせよ。

(1) $B + C$

答. $4x^2 - 6x - 9$

(2) $A - 3C$

答. $-3x^2 + 10x + 18$

(3) $A - B - C$

答. $-x^2 + 10x + 12$

(4) $-A + B - C$

答. $-3x^2 - 6x - 2$

(5) $-A - 2B + 3C$

答. $-x^2 - 2x - 10$

(6) $-3A - 2B + C$

答. $-11x^2 - 6x - 6$

(7) $A + 2B - 3C$

答. $x^2 + 2x + 10$

(8) $2A - 3B + 3C$

答. $6x^2 + 14x + 3$

2. 次の1次方程式を解け。

(1) $5x - 5 = 4x - 5$

答. $x = 0$

(2) $5x - 4 = -2x + 3$

$7x = 7$

答. $x = 1$

(3) $-3x + 3(6x - 7) = 5$

$-3x + 18x - 21 = 5$

$15x = 26$

答. $x = \frac{26}{15}$

(4) $5x + 8 - (4x + 4) = 0$

$5x + 8 - 4x - 4 = 0$

$x = -4$

答. $x = -4$

(5) $2x - (5x + 6) = 4$

$2x - 5x - 6 = 4$

$-3x = 10$

答. $x = -\frac{10}{3}$

(6) $-\frac{9}{4}x - 1 = 3x + 1$

両辺に4をかけて, $-9x - 4 = 12x + 4$

答. $x = -\frac{8}{21}$

(7) $\frac{9}{2}x - 9 = -6x + \frac{7}{4}$

両辺に4をかけて, $18x - 36 = -24x + 7$

答. $x = \frac{43}{42}$

(8) $-5x - \frac{1}{4} = 2x + \frac{7}{4}$

両辺に4をかけて, $-20x - 1 = 8x + 7$

答. $x = -\frac{2}{7}$